Numerical Modelling using Metropolis Algorithm

Results: -

By running the python simulation with N=20000 iterations at T=300K, I was able to successfully extract the following parameters which matched with the values collected from literature

Average Potential Energy: 2.0858e-21 J

Theoretical Energy (0.5kT): 2.0710e-21 J

Heat Capacity (Cv): 7.3601e-24 J/K

Summary: -

Numerical Integration: Using Python's scipy.integrate library, we calculated the exact mathematical integral for the average energy. The values matched with the maths.

Error Analysis: We calculated the fractional error of our simulation for different numbers of steps (N=10, 100, 1000...). As expected by statistical theory, plotting these results proved that the error drops reliably as $1 / \sqrt{n}$

Project successfully demonstrates how Python and stochastic numerical modeling (Monte Carlo methods) can be used to bypass impossible calculus and accurately predict the physical and thermal behavior of complex systems.